Chunk Size and Overlap in a FAISS-Based Retrieval Pipeline: A Controlled Comparison of Twelve Configurations over Philippine SEC Online-Lending Records

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***Abstract*—Chunk size and overlap decide which passages a retrieval-augmented generation system can return, and both are usually set by convention. This paper compares twelve fixed-size chunking configurations, a 4x3 factorial of chunk size (300, 500, 1000 and 1800 characters) and overlap ratio (0, 10 and 20 percent), on a 126-page corpus of prepared summaries of Philippine Securities and Exchange Commission records on online lending platforms. Nine questions at three difficulty levels were broken into eighteen evidence units, each located by its exact text before any configuration was run. Every configuration was embedded with all-MiniLM-L6-v2, indexed in a newly built FAISS IndexFlatIP and evaluated at k=3. The three 1800-character configurations tie for the best scores (Precision@3 0.444, Recall@3 0.685, MRR@3 0.667), because every cleaned page fits in one chunk and overlap has no effect at that size. Exact sign tests separate them from only one of the other eleven configurations, and no comparison survives Bonferroni correction, so the ranking from 500 characters up is not settled. Two results are better supported. Scoring relevance by character overlap instead of full containment raises Recall@3 by 0.105 at 300 characters and by nothing at 1000 or 1800, so the relevance rule interacts with the variable under test. Every configuration below 1000 characters also leaves some evidence that no chunk holds whole, with the reachable share falling to 0.796 at 300 characters with 30 of overlap. The largest remaining failure is lexical: questions that name products such as Peso2you and Mabilis Cash were not answered at any chunk size.**

***Index Terms*—Document retrieval, FAISS, information retrieval, Mean Reciprocal Rank, Precision@3, Recall@3, retrieval-augmented generation, text chunking, vector search.**

I. INTRODUCTION

A retrieval-augmented generation (RAG) system answers a question from passages retrieved from a document collection instead of relying only on the parameters of a language model [1]. Its output is bounded by retrieval, since an answer cannot be grounded in evidence the retriever never supplies. Before retrieval can happen, a document must be divided into passages, and the most common strategy is fixed-size character chunking, which has two free parameters. The chunk size sets how much context comes with each candidate passage, and the overlap decides whether a passage cut by one boundary is recovered by its neighbour.

These parameters are often set by convention. The baseline supplied for this exercise, 1000 characters with 200 characters of overlap, is one such convention. Recent work finds that no single setting is best everywhere: Bhat et al. report that the best chunk size depends on both the dataset and the embedding model, with concise fact-based collections favouring small chunks and context-dependent collections favouring large ones [4]. The setting therefore has to be measured on the corpus a system will serve.

This paper reports that measurement for the corpus behind our group's question-answering application. It compares twelve chunking configurations over prepared SEC lending records, scored against evidence units fixed in advance with three retrieval metrics. It also reports two measurement effects that apply beyond this corpus: the rule used to judge a retrieved chunk relevant changes its verdicts with chunk size, and configurations below 1000 characters cannot reach some evidence at any rank. The result is a configuration recommendation for this corpus and the evidence behind it.

II. RELATED WORK

A. Passage granularity in information retrieval

The question of what unit to retrieve is older than dense retrieval. Callan showed that scoring fixed-length passages instead of whole documents improves retrieval effectiveness, and that passage boundaries interact with document structure [3]. The present experiment asks the same question in a modern setting: the passage unit is set by a character budget, and similarity is computed in a learned embedding space.

B. Dense retrieval and vector indexing

Dense retrieval replaces lexical matching with similarity between learned vectors. Karpukhin et al. showed that a dual encoder trained on question-passage pairs outperforms strong lexical baselines on open-domain question answering [2], and Reimers and Gurevych introduced the sentence-embedding architecture from which the model used here is derived [6]. At retrieval time the vectors are searched by a similarity index, and FAISS provides both exact and approximate implementations [7]. This study uses an exact index so that no approximation error is mixed into the effect of chunking.

C. Chunking strategies for RAG

Chunking has recently been studied directly. Bhat et al. examine fixed-size chunking across several datasets and embedding models and find no universally best size, reporting that chunks of 64 to 128 tokens suit concise factual answers while 512 to 1024 tokens suit context-dependent ones, with the preferred size varying by embedding model [4]. Merola and Singh compare fixed-size chunking with late chunking and

contextual retrieval, and report that context-preserving methods improve semantic coherence at extra computational cost [5].

The present study is narrower than either. It measures the two parameters of the fixed-size strategy used in our pipeline, on the corpus that pipeline will serve, and it does not compare against late or contextual chunking. Its results agree with the finding of [4] that the preferred configuration depends on the corpus, and it adds a measurement point that, to the author's knowledge, that literature does not address: the relevance rule itself interacts with chunk size.

III. METHODOLOGY

A. Dataset

The corpus is the Member 3 file of the LendSuri PT-F1 single-PDF package [8]. It has 126 pages, and each page is one prepared summary of a Philippine Securities and Exchange Commission (SEC) record on lending companies and online lending platforms (OLPs) [9]. Pages 1 to 60 come from the January 26, 2026 registry snapshot of recorded OLPs, pages 61 to 101 from the November 29, 2021 historical list, pages 102 to 123 from SEC regulatory events and advisories, pages 124 and 125 from framework guidance, and page 126 from complaint guidance. It was used because it covers the domain of the group's planned question-answering system on Philippine online lending.

Three properties make the corpus a hard retrieval target. Every page follows the same record template, so most of the text on any page is shared with every other page, and only the platform name, company name, dates and advisory title tell records apart. Many platform names are built from the same short words, such as Peso, Cash and Loan, so a query that names one product shares those subwords with dozens of unrelated records. The same company often appears on several pages, for example in the 2026 snapshot, the 2021 list and a later advisory, so finding the right company does not guarantee finding the right record.

B. Preprocessing

Every page has a text layer and pypdf extracts all 126 pages, so no optical character recognition was needed. The package file does repeat four elements on every page: a running header, a printed page number taken from the original category PDF, the label "LendSuri / Member 3" and a two-line disclaimer footer. Together they come to 34,368 of 164,290 characters, or 20.9 percent of the corpus.

These elements were removed. At 300 characters they fill much of a chunk, while at 1800 they are a small share of a page, so leaving them in would favour large chunks for reasons unrelated to retrieval. The printed page number would also have put a wrong page reference into the indexed text. The remaining lines were re-rendered one record per page, so each record keeps its page number and the package's gold pages apply unchanged. The cleaned file holds 129,983 characters after the pipeline's whitespace normalisation and was used unchanged throughout; the original file is kept as a control. The answer key, question sheet and page map that come with the package were never indexed.

C. Chunking configurations

Twelve configurations were tested, formed as a 4x3 factorial of chunk size and overlap ratio. Chunk sizes of 300, 500, 1000 and 1800 characters were each paired with overlaps of 0, 10 and 20 percent of the chunk size. Overlap is expressed as a ratio so that sizes can be compared at equal redundancy; a fixed 200-character overlap would mean 67 percent redundancy at 300 characters and 11 percent at 1800. Chunking runs separately within each page, so no chunk spans a page boundary and each page contributes a shorter final chunk. A cleaned page is at most 1,444 characters long and about 1,030 on average, so at 1800 characters every page becomes a single chunk. Configuration C9 reproduces the 1000/200 baseline.

D. Retrieval pipeline and controls

The dataset, embedding model, vector store, retrieval depth, test questions and relevance criteria were held constant across all twelve configurations. Embeddings were produced by all-MiniLM-L6-v2 [6], which yields 384-dimensional vectors, normalised to unit length so that inner product equals cosine similarity. Retrieval used a FAISS IndexFlatIP [7], which searches exhaustively and returns exact nearest neighbours. Retrieval depth was fixed at k=3 to match the metrics. A new index was built for each configuration and no index was reused.

One limitation of the embedding model is recorded and left in place. all-MiniLM-L6-v2 truncates input at 256 word-pieces, and at 1800 characters 69 of the 126 cleaned pages are longer than that, so the end of those pages is not embedded although the whole page is returned when retrieved. None of the eighteen evidence passages falls in a truncated part, so no gold evidence is lost from the embeddings. Section VI returns to this point.

E. Test questions and evidence units

Nine questions were used, three at each difficulty level in the exercise brief: easy questions answerable from one explicit passage, average questions needing interpretation or closely related facts, and difficult questions needing evidence from different records or the separation of similar names. Three of them, M3-E, M3-A and M3-D, are the Member 3 questions supplied with the package, with the answer key's gold pages. The other six were written from the corpus to cover record types the first three do not use: the complaint guidance (E3), the framework guidance (A3) and the 2021 historical list (A2, D2 and D3). Their answers were checked against the corpus before any configuration was run.

Each question was broken into evidence units, eighteen in total. An evidence unit is the shortest passage on a gold page that supports one part of the expected answer, such as "Platform: Peso2you. Company: Snapcash Lending, Inc." on page 46. Units are stored as exact text and located in the pipeline's whitespace-normalised page text when the notebook

runs, and the run stops if a unit is missing or occurs more than once. Because a unit is defined by its position in the page, the same evidence is tracked across all twelve configurations even though chunk numbering changes with each one, and the gold chunks are re-derived inside the experiment loop.

F. Relevance criterion

A retrieved chunk counts as supporting an evidence unit only when it contains that unit in full. The exercise brief requires that a relevant chunk contain evidence that supports the answer, and states that mentioning the same topic or keywords is not enough. On this corpus the rule matters more than usual, because a chunk from the right page can hold nothing but template text that every page shares. Under an overlap rule, a chunk holding the last few characters of a 51-character registry line would still count as relevant. Section V-A reports what this choice changes, and the effect differs by chunk size.

G. Evaluation metrics

Let R be the ordered list of k=3 retrieved chunks for a question, rel(c) = 1 when chunk c contains at least one evidence unit in full and 0 otherwise, U the set of evidence units the question requires, and U_cov the subset of U for which some chunk in R contains the unit in full. Precision@3 is given by (1), Recall@3 by (2) and the reciprocal rank by (3), where rank(R) is the position of the first chunk with rel(c) = 1; RR is 0 when no chunk in R is relevant.

$$P@3 = (1/3) \Sigma\, rel(c),\ \ c \in R \tag{1}$$

$$R@3 = | U_cov | / | U | \tag{2}$$

$$RR = 1 / rank(R) \tag{3}$$

Reciprocal rank is therefore 1.000 when the first retrieved chunk is relevant, 0.500 at the second, 0.333 at the third and 0.000 when no relevant chunk appears. MRR@3 is the mean of RR over all nine questions, as in (4). It is not computed by averaging the three difficulty-level means, since those groups could in general differ in size.

$$MRR@3 = (1/Q) \Sigma\, RR_q\ ,\ \ Q = 9 \tag{4}$$

Defining Recall@3 over evidence units instead of gold chunks fixes the denominator of (2) for each question, so it does not drift with chunk size. A single chunk may count toward every unit it contains, and a unit found in two retrieved chunks counts once.

A fourth quantity is reported with these. The maximum achievable Recall@3 in (5) is the share of a question's evidence units that some chunk in the configuration contains in full, whether or not retrieval returns it, written U_reach. When it falls below 1 the configuration cannot supply the missing evidence at any rank, and the shortfall is a property of the segmentation.

$$Max\ R@3 = | U_reach | / | U | \tag{5}$$

H. Statistical treatment

With nine questions, one relevance decision moves Precision@3 by 0.037, so differences of that size carry little information. Two analyses are therefore reported with the point estimates. Percentile bootstrap confidence intervals were computed by resampling the nine questions with replacement over 10,000 replicates. Pairwise comparisons against the leading configuration used the exact two-sided sign test on per-question Recall@3, with tied questions excluded, and are reported both uncorrected and against a Bonferroni threshold for the eleven comparisons.

I. Efficiency measurement

Chunk count and index size are exact, since they follow from the segmentation and the embedding dimension. Embedding time, index construction time and mean query latency were recorded from single runs on a shared Colab T4 runtime and are reported for completeness only. They are not controlled measurements, for the reasons in Section VI-D.

J. Reproducibility

The experiment is contained in the notebook submitted with this report, which holds the chunking, embedding, indexing and scoring code in the order executed, together with the nine questions and the exact text of their eighteen evidence units. The accompanying workbook holds the configuration-level results of Table I, the difficulty breakdown of Table II, the per-question scores and the full retrieval log of 324 records, one per configuration, question and rank. Retrieval is deterministic, since the embedding checkpoint is fixed and IndexFlatIP searches exhaustively, so rerunning the notebook on the cleaned corpus reproduces Table I. The experiment was run twice, once on the Colab GPU runtime and once on a local CPU, and produced identical metrics. Only the bootstrap intervals change between runs, because they are resampled. The recommended configuration is reproduced by setting a chunk size of 1800 characters and an overlap of 180 characters at k=3.

IV. RESULTS

The experiment produced 324 retrieval records, one per configuration, question and rank. Table I gives the configuration-level results, and Fig. 1 adds bootstrap confidence intervals. Index size is not tabulated because it follows directly from chunk count: each vector holds 384 single-precision values, or 1.5 KB per chunk.

TABLE I

CONFIGURATION-LEVEL RETRIEVAL RESULTS. Max R@3, from (5), is the highest Recall@3 the segmentation allows regardless of retrieval quality.

Cfg	Size	Ovl	Ratio	Chunks	P@3	R@3	MRR	Max R
C1	300	0	0%	475	0.222	0.296	0.185	0.926
C2	300	30	10%	558	0.111	0.185	0.148	0.796
C3	300	60	20%	600	0.185	0.259	0.148	0.926
C4	500	0	0%	333	0.111	0.259	0.148	0.833
C5	500	50	10%	339	0.222	0.389	0.426	0.870
C6	500	100	20%	374	0.370	0.519	0.556	0.944
C7	1000	0	0%	207	0.407	0.574	0.611	1.000
C8	**1000**	**100**	**10%**	**210**	**0.370**	**0.519**	**0.500**	**1.000**
C9	1000	200	20%	235	0.407	0.574	0.611	1.000
C10	1800	0	0%	126	0.444	0.685	0.667	1.000
C11	1800	180	10%	126	0.444	0.685	0.667	1.000
C12	1800	360	20%	127	0.444	0.685	0.667	1.000

C10, C11 and C12, at 1800 characters, share the highest value on all three metrics: Precision@3 of 0.444, Recall@3 of 0.685 and MRR@3 of 0.667. Their results are identical because no cleaned page is longer than 1,444 characters, so each page is one chunk whatever the overlap; C12 differs only by one extra four-character chunk. The 1000-character configurations follow with Recall@3 between 0.519 and 0.574. Six configurations, C7 to C12, reach all eighteen evidence units, and the lowest ceiling is C2 at 0.796.

A. Performance by difficulty

The difficulty levels do not order the results the way the labels suggest (Table II). Difficult questions reach Recall@3 of 0.889 in every configuration from 1000 characters up, while easy questions never pass 0.667 and average questions never pass 0.500.

TABLE II

RECALL@3 AND MRR@3 BY QUESTION DIFFICULTY. Precision@3 by difficulty is in the submitted workbook; it is omitted here because it is not comparable across levels, as explained below.

			Easy		Average		Difficult	
Cfg	Size	Ovl	R@3	MRR	R@3	MRR	R@3	MRR
C1	300	0	0.333	0.111	0.000	0.000	0.556	0.444
C2	300	30	0.333	0.111	0.000	0.000	0.222	0.333
C3	300	60	0.333	0.111	0.000	0.000	0.444	0.333
C4	500	0	0.667	0.333	0.000	0.000	0.111	0.111
C5	500	50	0.667	0.500	0.167	0.333	0.333	0.444
C6	500	100	0.667	0.333	0.333	0.333	0.556	1.000
C7	1000	0	0.667	0.500	0.167	0.333	0.889	1.000
C8	**1000**	**100**	**0.667**	**0.500**	**0.000**	**0.000**	**0.889**	**1.000**
C9	1000	200	0.667	0.500	0.167	0.333	0.889	1.000
C10	1800	0	0.667	0.500	0.500	0.500	0.889	1.000
C11	1800	180	0.667	0.500	0.500	0.500	0.889	1.000
C12	1800	360	0.667	0.500	0.500	0.500	0.889	1.000

Precision@3 has a structural ceiling. A question with one evidence unit usually allows only one relevant chunk among three, so its Precision@3 rarely exceeds 0.333. It can go higher only when overlapping chunks each hold the whole unit, which happened once (E2 at C6, 0.667). All three easy questions need one unit each, the average questions need two and the difficult questions need three, so easy questions score lower on precision for reasons unrelated to retrieval quality.

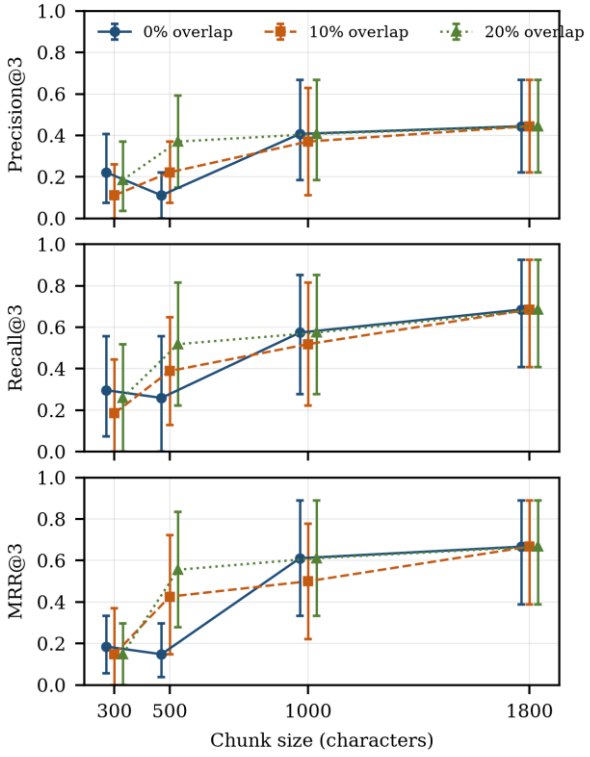


Fig. 1. The three retrieval metrics against chunk size for all twelve configurations, one line per overlap ratio. Points are configuration means with percentile bootstrap 95 percent confidence intervals over the nine questions (10,000 resamples), offset horizontally to avoid occlusion. All three metrics rise with chunk size, and the three overlap lines meet at 1800 characters, where every page fits in one chunk. The intervals overlap widely, which is the basis for the result in Section IV-B. Precision@3 is subject to the per-question ceiling described in Section IV-A. Exact values and the Max Recall@3 ceiling are in Table I.

B. Statistical separation

The point estimates are not well separated. C10 records Recall@3 of 0.685 with a bootstrap interval of [0.389, 0.944], against [0.278, 0.852] for C7 and C9 and [0.222, 0.815] for C8. Exact two-sided sign tests on per-question Recall@3, excluding ties, separate C10 only from C2 ($p = 0.031$). C1, C3 and C4 are marginal at $p = 0.063$, with C10 higher on five questions and lower on none. C10 differs from C7 and C9 on one question and from C8 on two, giving $p = 1.000$ and $p = 0.500$. The Bonferroni threshold for eleven comparisons is 0.0045, which no comparison meets. The experiment therefore does not show that 1800-character chunks beat 1000-character chunks on this corpus, and this limits the recommendation in Section VII.

V. DISCUSSION

A. The relevance criterion interacts with chunk size

Re-scoring the same retrieval output with a looser rule, under which a chunk counts when it overlaps an evidence unit by any number of characters, raises mean Recall@3 by 0.105 at 300 characters and by 0.099 at 500, and leaves it unchanged at 1000 and 1800. The gradient follows from chunk length. A

small chunk often clips a record line and registers a partial match, while a chunk of 1000 characters or more holds a whole line or none of it, which leaves nothing for the looser rule to credit.

Under the looser rule C6 would score 0.630 instead of 0.519 and move above every 1000-character configuration. A comparison of chunk sizes should therefore state its relevance rule, because that rule is not independent of the variable being tested. The chunking studies surveyed here [4], [5] evaluate against relevance judgments but do not report this interaction.

B. Evidence that no chunk can supply

Full containment also exposes evidence that no chunk can supply. In every configuration below 1000 characters some evidence units straddle a chunk boundary, so retrieval cannot recover them at any rank: C2 reaches thirteen of the eighteen units and has a Max Recall@3 of 0.796, and C4 reaches fourteen. From 1000 characters up every unit is reachable. This is the fragmentation problem that motivates context-preserving methods [5], and it agrees with Callan's observation that passage boundaries interact with document structure [3]. Evidence completeness should be reported apart from the retrieval metrics.

M3-E shows the mechanism at the boundary. Its only unit, "Platform: Peso2you. Company: Snapcash Lending, Inc.", sits at characters 390 to 441 of page 46. At C2 the retriever did return a chunk from page 46, but that chunk spans characters 810 to 1110 and holds only template text, so it is correctly scored as a miss. It is the only time any configuration returned page 46 for this question.

C. Difficulty labels do not predict retrievability

Mean Recall@3 per question across the twelve configurations runs from 1.000 for E3 to 0.000 for M3-E and A2, and the order does not follow the difficulty labels. M3-E is an easy single-unit question and was never answered, while D2, a difficult question with three units, averages 0.750 and is fully answered in seven configurations. The labels describe the reasoning a reader must do, which is a different property from how easily a dense retriever finds the text.

D. Product names that share common subwords

The two questions no configuration answered both ask about a product name. For M3-E the top-ranked chunk is page 99, the 2021 record of PesoQ under U-Peso Lending Investors Corp., in ten of the twelve configurations, and the other retrieved pages are also Peso-named records such as Pesopop and PesoCow (page 88) and PESO CASH (page 121). For A2, about Mabilis Cash, the retriever returns Cash-named records instead, such as Cashme and Cashwow (page 76) and Cash-Express (pages 3 and 65). E2 shows a milder form: the 2021 Atome PH record (page 84) often outranks the January 2026 record (page 35) the question asks about. The difficult questions name distinctive companies, Tala and ACOM, and use the advisories' own words (Facebook page, app, advisory), and they are answered well. Chunk size does not fix the product-name failures, since both questions' evidence is reachable from 1000 characters up. A retriever that also matches exact names, such as a hybrid of lexical and dense search, is the likely fix and is outside this study.

E. Relation to prior findings

The result agrees with [4] that the best chunk size depends on the dataset and the embedding model. On this corpus the point estimates favour whole-record chunks, which suits a collection of short, self-contained records. The comparison should not be taken further: [4] measures in tokens across several public datasets and models, this study measures characters on one corpus with one model, and its intervals do not separate the candidates.

F. Retrieval quality against index cost

Each chunk adds one 384-dimensional single-precision vector of 1.5 KB. The twelve configurations span 126 to 600 chunks, or 189 to 900 KB. On this corpus a larger index does not buy better retrieval: the 300-character configurations build the largest indexes, 475 to 600 chunks, and are among the weakest on Recall@3, while the 1800-character configurations build the smallest, 126 or 127 chunks, and record the highest. C10 and C11 use the same 126 chunks, so the 10 percent overlap of C11 costs nothing here. The timing figures are left out for the reason given in Section VI-D.

VI. LIMITATIONS

A. Statistical power

Nine questions and eighteen evidence units are enough to show that the smallest chunks do worse, at least for C2, but not to order the configurations from 500 characters up; no pairwise difference survives correction. Part 1 of this exercise, using only the three package questions, found 1000/0, 1000/200 and 1800/200 tied on all three metrics, which shows again how little a small test set can separate.

B. The relevance criterion

Full containment is strict and sometimes misfires. At C2 the rank-1 chunk for M3-D holds 143 of the 150 characters of the June 9, 2026 Tala advisory and misses only the leading label "Source/", yet it is scored a miss. A threshold rule such as 80 percent containment would score it a hit. The rule was fixed before the results were examined and was not revised afterwards, because changing a metric in response to the scores it produces would invalidate the comparison.

C. Dataset, model and scope

The corpus is 126 prepared summaries in one fixed template, drawn from one package, and the records are summaries of SEC sources, not the original SEC documents [8]. The conclusions may not carry over to prose documents, to raw SEC PDFs or to collections in which records vary in length. Removing the running header and footer changed every configuration in the same way, so it does not bias the comparison, but results on the uncleaned file would differ.

One embedding model was tested, and [4] reports that the preferred chunk size varies by model, so the result is specific to all-MiniLM-L6-v2 on this corpus. That model truncates at 256 word-pieces, and 69 of the 126 pages are longer than that, so at 1800 characters those chunks are matched on their opening part while the whole page is returned. Only fixed-size character chunking was evaluated, within a 4x3 grid, at a fixed retrieval depth of k=3.

D. Efficiency measurements

Chunk count and index size are exact. The timing figures are not controlled measurements. Each was recorded once on a shared Colab T4 runtime without repetition, embedding times of 0.39 to 0.53 seconds are too close to separate, and mean query latency of about 7 ms below 1000 characters against 9 to 11 ms above it comes from single runs. No conclusion rests on these figures.

VII. CONCLUSION AND FUTURE WORK

Twelve fixed-size chunking configurations were compared for a FAISS retrieval pipeline over prepared SEC records on Philippine online lending, each scored against eighteen evidence units fixed before the runs.

The experiment does not establish a single best configuration. Of eleven pairwise comparisons against the leading configuration, one reaches significance uncorrected and none survives Bonferroni correction.

Three directional results are supported. The five lowest Recall@3 values all belong to configurations below 1000 characters, and C2 differs from the leader at $p = 0.031$. Every configuration below 1000 characters leaves some evidence unreachable at any rank, down to a Max Recall@3 of 0.796 for C2, while every configuration from 1000 characters up reaches all eighteen units. Questions that name a product built from common subwords, Peso2you and Mabilis Cash, were not answered by any configuration, while the difficult questions about distinctive companies reached Recall@3 of 0.889.

On the evidence available, the recommended configuration is 1800 characters with 180 of overlap, C11. It shares the highest Precision@3, Recall@3 and MRR@3 and the best difficult-question result, reaches all eighteen evidence units and uses the smallest index, 126 chunks. On this corpus it behaves exactly like C10, because no page is longer than 1,444 characters; the 10 percent overlap is kept because it costs nothing here and would help any future record longer than one chunk. The main caution is the embedding window. Since 69 of the 126 pages are longer than 256 word-pieces, C9 (1000 characters with 200 of overlap) is the fallback if later records place answer text near the end of a page, because its shorter chunks are less affected by truncation and its scores are not statistically separable from those of C11.

Three extensions would strengthen these findings: a test set of thirty or more questions, to gain the power that nine questions lack; an embedding model with a longer input window, to test the 1800-character result without truncation; and a hybrid retriever that adds exact name matching to dense retrieval, since no chunk size fixed the product-name failures.

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REFERENCES

[1] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 9459–9474.

[2] V. Karpukhin et al., "Dense passage retrieval for open-domain question answering," in Proc. 2020 Conf. Empirical Methods in Natural Language Processing (EMNLP), Online, Nov. 2020, pp. 6769–6781, doi: 10.18653/v1/2020.emnlp-main.550.

[3] J. P. Callan, "Passage-level evidence in document retrieval," in SIGIR '94, B. W. Croft and C. J. van Rijsbergen, Eds. London, U.K.: Springer, 1994, pp. 302–310, doi: 10.1007/978-1-4471-2099-5_31.

[4] S. R. Bhat, M. Rudat, J. Spiekermann, and N. Flores-Herr, "Rethinking chunk size for long-document retrieval: A multi-dataset analysis," arXiv:2505.21700, May 2025.

[5] C. Merola and J. Singh, "Reconstructing context: Evaluating advanced chunking strategies for retrieval-augmented generation," arXiv:2504.19754, Apr. 2025.

[6] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. 2019 Conf. Empirical Methods in Natural Language Processing and 9th Int. Joint Conf. Natural Language Processing (EMNLP-IJCNLP), Hong Kong, China, Nov. 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.

[7] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," IEEE Trans. Big Data, vol. 7, no. 3, pp. 535–547, Jul. 2021, doi: 10.1109/TBDATA.2019.2921572.

[8] "LendSuri PT-F1 single-PDF student package: Member 3 corpus," prepared research summaries of SEC lending records based on the LendSuri v5 corpus, Oct. 23, 2026.

[9] Securities and Exchange Commission, "List of recorded online lending platforms," Manila, Philippines. [Online]. Available: https://www.sec.gov.ph/lending-companies-and-financing-companies-2/list-of-recorded-online-lending-platforms/